

SATHWIK H NAIK

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PROFESSIONAL SUMMARY

Recent M.S. Data Science graduate (University of Maryland, College Park, 2026, GPA 3.81/4.0) with hands-on experience spanning data analytics, data engineering, data science, and AI/ML engineering. Built Power BI dashboards, ETL pipelines, and workflow automation at a Fortune 50 healthcare organization. Completed seven self-directed capstones covering the full modern data stack, including Kafka/Spark/dbt/Snowflake lakehouse pipelines, SQL and BI analytics, RAG system evaluation, banking forecasting and causal inference, and generative and small-language-model research.

EDUCATION

University of Maryland, College Park
M.S. in Data Science

2024 – 2026
GPA 3.81/4.0

Coursework Statistical Inference, Deep Learning, Data Modeling, Data Visualization, Experimental Design, Advanced Time Series

Reva University, India
B.Tech, Computer Science and Engineering

2019 – 2023
GPA 3.55/4.0

WORK EXPERIENCE

Elevance Health

Aug 2023 – Aug 2024

Software Engineer

Tech Stack: Salesforce CRM, Informatica, Bitbucket, Power BI, DAX, AutoRABIT, Apex

- Designed reusable KPI structures, translating operational requirements into consistent, self-serve cross-team metrics
- Built and maintained 10+ Power BI dashboards with DAX measures for multiple internal teams
- Developed ETL workflows to source, clean, and validate healthcare data, improving quality for stakeholder reports
- Automated recurring data-preparation and reporting workflows, eliminating 20+ hours of manual weekly effort
- Documented report logic, metric definitions, and validation rules enabling governed self-service reporting
- Supported CI/CD pipelines for Salesforce releases using automated logging and standardized deployment workflows

PROJECTS

Healthcare Insurance Claims Analytics & Fraud Signal Platform ([GitHub](#))

Tech Stack: Kafka (Redpanda), Spark (Structured Streaming), Delta Lake/Databricks, dbt, Airflow, Snowflake, Terraform

- Built a hybrid lakehouse-warehouse pipeline using Kafka, Spark, Delta Lake, dbt, Airflow, and Snowflake
- Implemented Bronze/Silver/Gold medallion architecture on Delta Lake processing 50,000 synthetic insurance claims
- Built two dbt projects with tests, transforming raw claims into governed, BI-ready Snowflake marts
- Provisioned Databricks and Snowflake infrastructure declaratively using Terraform with key-pair JWT authentication
- Built a rolling-baseline fraud detection rule in dbt achieving 100% recall and 67% precision
- Built Spark Structured Streaming ingestion using explicit schemas, checkpointing, and micro-batch triggers

SaaS Subscription Churn & Revenue Analytics Platform ([GitHub](#))

Tech Stack: DuckDB, pandas, PostgreSQL, Grafana, Plotly, scikit-learn, SHAP

- Built a SaaS churn analytics platform across DuckDB, pandas, PostgreSQL, Grafana, and scikit-learn
- Modeled the MRR waterfall and running balance in SQL using CTEs and window functions
- Reconstructed point-in-time customer state using pandas `merge_asof` across 8,000 synthetic customer accounts
- Derived 15+ metrics tables including NRR, GRR, and ARPA from raw subscription events
- Built a 21-panel Grafana dashboard provisioned entirely as version-controlled config, not UI clicks
- Trained a Gradient Boosting churn model and explained predictions per-customer using SHAP

RAG Evaluation Lab ("Retrieval Quality Lab") ([GitHub](#))

Tech Stack: Python, ChromaDB, Ollama, RAGAS, LangChain, BM25, sentence-transformers

- Built a RAG evaluation pipeline comparing BM25, dense, hybrid, and reranked retrieval methods
- Achieved 85.3% Recall@1 with hybrid retrieval plus cross-encoder reranking, up from 69.6% BM25
- Evaluated generation quality using EM, F1, a custom grounding score, and RAGAS relevancy
- Diagnosed generation, not retrieval, as the system bottleneck despite 95% gold-doc accuracy
- Built a 5-tab Streamlit dashboard visualizing retrieval ablations, generation quality, and latency
- Ran a local Ollama LLM for generation and LLM-as-judge evaluation with no paid APIs

Banking Liquidity Forecast Lab ([GitHub](#))

Tech Stack: Python, LightGBM, statsmodels, FastAPI, Docker Compose

- Built a retail-banking cash-outflow forecasting pipeline comparing seasonal naive, Holt-Winters, and LightGBM models
- Achieved 9.48% WAPE forecasting daily cash outflow with Holt-Winters under six-fold walk-forward validation
- Built a residual-based anomaly detector using robust z-scores, reaching 0.71 F1 at threshold
- Measured a marketing campaign's causal lift using interrupted time series and difference-in-differences
- Recovered a known 12% campaign lift within 0.4 percentage points using two causal estimators
- Served the forecasting model through FastAPI endpoints packaged in reproducible Docker Compose containers

Conditional Generative Data Augmentation for Image Classification ([GitHub](#))

Tech Stack: Python, PyTorch, Conditional GAN (cGAN), Conditional VAE-GAN, CNN, MNIST, Fashion-MNIST

- Developed CGAN and Conditional VAE-GAN pipelines in PyTorch across MNIST and Fashion-MNIST
- Improved Fashion-MNIST accuracy from 63.6% to 70.8% using only 100 real samples
- Evaluated seven classifier training strategies, finding weighted real-synthetic mixing performed best overall
- Ran architecture and training-strategy ablations, finding learning-rate ratio the strongest performance lever
- Found VAE-GAN augmentation underperformed real-only training, showing complexity alone doesn't improve results
- Identified a large-latent VAE-GAN variant as the only architecture improving both dataset's F1

TinyACE: Bounded Working-Memory Context Engineering for SLMs ([GitHub](#))

Tech Stack: Python, PyTorch, TinyLlama-1.1B, Phi-3 Mini (3.8B), Mistral-7B, sentence-transformers

- Developed TINYACE, a bounded working-memory framework capping context at 512 tokens for small models
- Improved Phi-3 Mini accuracy from 74% to 78% across a 50-question SciQ benchmark
- Ran a multi-scale evaluation across TinyLlama, Phi-3 Mini, and Mistral on one GPU
- Identified failure tracking as the most critical component through systematic ablation studies
- Found simple FIFO eviction matched or beat complex utility scoring for small models
- Diagnosed a capacity collapse where TINYACE degraded the smallest 1.1B model's accuracy

Advanced Time Series Analysis & Predictive Dashboarding

Tech Stack: Python, Docker, Git, Statistical Trend Analysis

- Built a refreshable ingestion pipeline monitoring rolling 24-hour windows of operational metrics
- Applied moving averages and trend signals to detect anomalies in high-frequency data
- Dockerized the workflow and used Git version control for reproducible execution across environments

TECHNICAL SKILLS

- **Data & SQL** SQL, ETL, Data Pipelines, Data Cleaning, Snowflake, DuckDB, PostgreSQL, Informatica
- **Data Engineering & Orchestration** dbt, Kafka (Redpanda), Spark (Structured Streaming), Airflow, Databricks (Delta Lake, Unity Catalog), Terraform, Docker Compose
- **LLM & RAG Engineering** RAG, Hybrid Search (BM25 + Dense + RRF), Cross-Encoder Reranking, ChromaDB, Ollama, RAGAS, LangChain, sentence-transformers, Hugging Face datasets
- **Statistics & Methods** Statistical Inference, Experimental Design, A/B Testing, Causal Inference (Interrupted Time Series, Difference-in-Differences), Time-Series Forecasting (Holt-Winters, Walk-Forward Validation), Model Evaluation
- **Programming & Tools** Python, pandas, NumPy, PyArrow, R, Git, Bitbucket, Docker, FastAPI, CI/CD
- **ML & AI** PyTorch, scikit-learn, LightGBM, Gradient Boosting, statsmodels, SHAP, GANs, VAE-GAN, CNNs, Deep Learning, LLM Context Engineering
- **Analytics & BI** Power BI, DAX, Tableau, Grafana, Plotly, Streamlit, Data Visualization
- **AI-Assisted Development** GitHub Copilot, Claude AI, Cursor, AI Debugging
- **Salesforce Ecosystem** Salesforce CRM, Apex, AutoRABIT